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MODULE 4

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SYLLUBUS

- Magnetic Measurements: Measurement of flux and permeability - flux meter, BH curve and permeability measurement - hysteresis measurement- ballistic galvanometer –principle- determination of BH curve - hysteresis loop. Lloyd Fisher square —measurement of iron losses.
- Measurement of luminous intensity-Photoconductive Transducers-Photovoltaic cells
- Temperature sensors-Resistance temperature detectors-negative temperature coefficient Thermistors - thermocouples-silicon temperature sensors.

INTRODUCTION

- Electrical and magnetic phenomena are closely associated with each other.
- The operating characteristics of electrical machines, apparatus and instruments are greatly influenced by the properties of ferro-magnetic materials used for their construction.
- Therefore, magnetic measurements and a thorough knowledge of characteristics of magnetic materials is of utmost importance in designing and manufacturing electrical equipment.

INTRODUCTION

- The principal requirements in magnetic measurements are :
 - (i) The measurement of magnetic field strength in air.
 - (ii) The determination of B-H curve and hysteresis loop for soft ferro -magnetic materials.
 - iii) The determination of eddy current and hysteresis losses of soft ferro -magnetic materials subjected to alternating magnetic fields.
 - (iv) The testing of permanent magnets.

MAGNETIC MEASUREMENTS AND TESTS

(i) BALLISTIC TESTS :- These tests are generally employed for the determination of B-H curves and hysteresis loops of ferro-magnetic materials. Direct current is used and a ballistic galvanometer or flux meter is employed for the measurement of flux density.

(ii) A. C. TESTING:- These tests may be carried at power, audio or radio frequencies. They give information about eddy current and hysteresis losses.

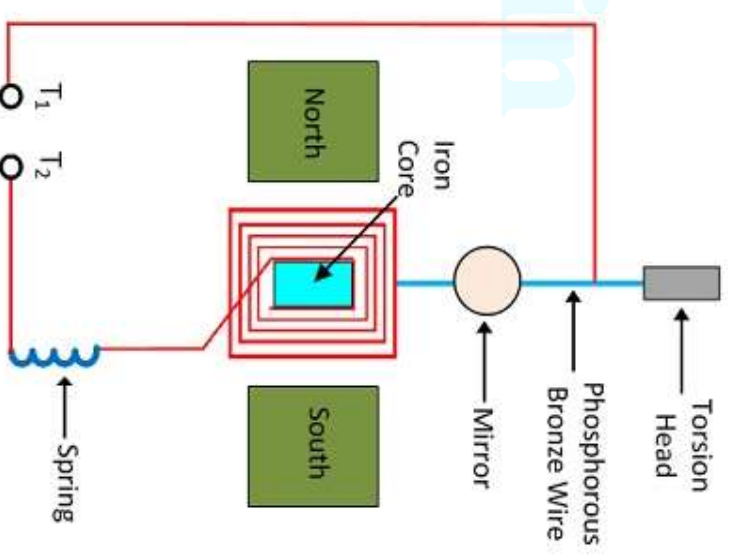
(iii) STEADY STATE TESTS : These are performed to obtain the steady value of flux density existing in the air gap of a magnetic circuit.

BALLISTIC GALVANOMETER

- The galvanometer which is used for estimating the quantity of charge flow through it is called the ballistic galvanometer.

➤ Construction of ballistic galvanometer

- The ballistic galvanometer consists of coil of copper wire which is wound on the non-conducting frame of the galvanometer.
- The phosphorous bronze suspends the coil between the north and south poles of a magnet.
- For increasing the magnetic flux the iron core places within the coil.
- The lower portion of the coil connects with the spring.
- This spring provides the restoring torque to the coil.



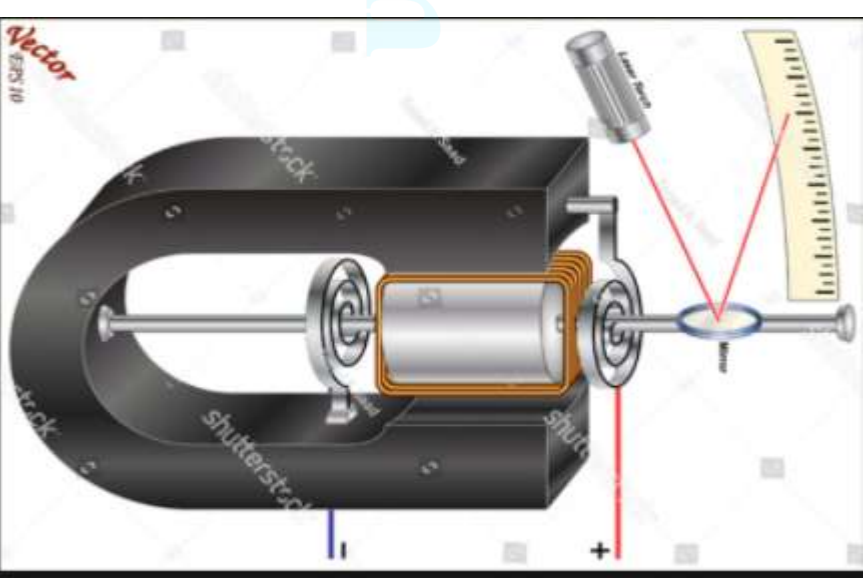
Ballistic Galvanometer

BALLISTIC GALVANOMETER

- The working principle of the ballistic galvanometer is very simple. It depends on the deflection of the coil which is directly proportional to the charge passes through it.
- Quantity of electricity in magnetic measurement is the e.m.f induced in a search coil on changing the magnetic flux linking the search coil.
- Search coil is a small coil in which a current can be induced to detect and measure a magnetic field.
- At equilibrium condition, we can write

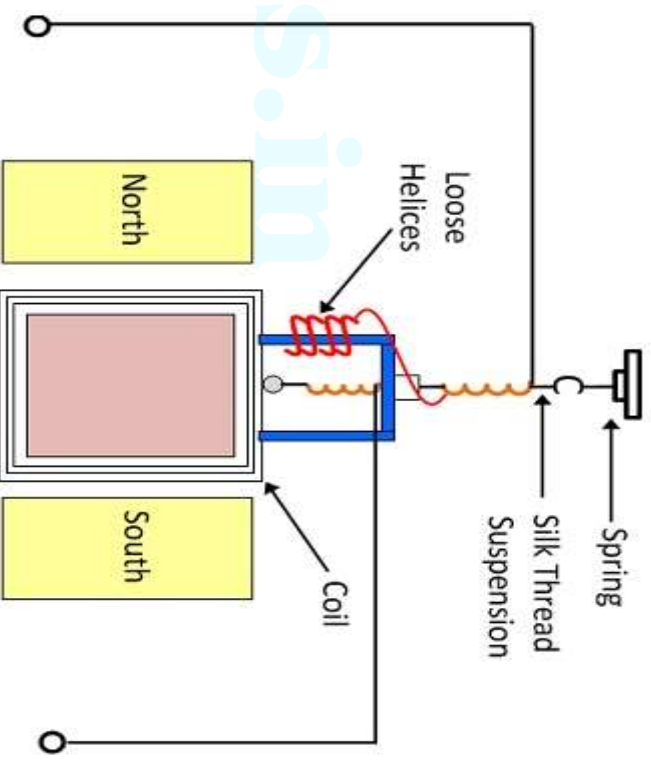
$$q = \frac{ct}{2\pi BNA} \times (\theta)$$

- The galvanometer measures the majority of the charge passes through it in spite of current.



FLUX METER

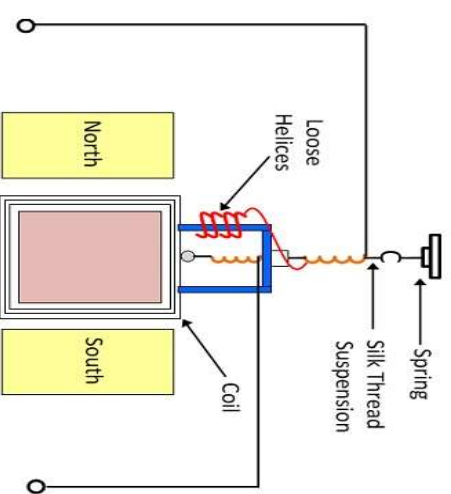
- The meter which is used for measuring the flux of the permanent magnet is known as the flux meter.
- The flux meter is the advanced form of the ballistic galvanometer and has certain advantages like the meter has low controlling torque and heavy electromagnetic damping.



FLUX METER

➤ Construction

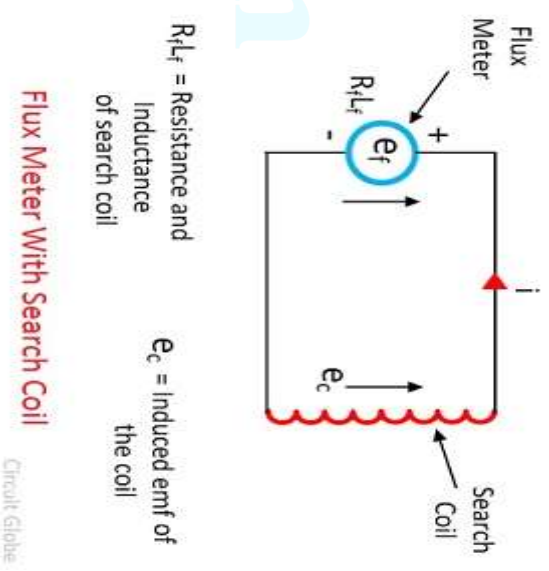
- The flux meter has a coil which is freely suspended by the help of the spring and the single silk thread. The coil moves freely between the poles of the permanent magnet.
- The current enters into the coil with the help of the helices which is very thin and made from the annealed silver strips.
- This current reduces the controlling torque to the minimum value.
- The air friction damping of the coil is negligible.



FLUX METER

➤ Operation

- The terminals of the fluxmeter are connected across the search coil as shown in the figure below.
- The flux linking with the coil is varied by either removing it from the magnetic field or by reversing the field of the magnet.
- The change of the flux induces the electromotive force in the coil.
- This emf induces the current in the search coil and send it through the flux meter.
- Because of the current, the pointer of the fluxmeter deflects, and their deflection is directly proportional to the change in the value of flux linkages.



FLUX METER

- As, the variation of the flux linkages reduces, coil stop moving because of their high electromagnetic damping.
- The high electromagnetic damping is because of the low resistance circuit between the fluxmeter and the search coil.

THEORY OF FLUX METER

- **R and L** are the total **resistance** and **inductance** respectively of the circuit comprising of the meter coil and search coil
- **N** is the **number of turns** on the **search coil**
- **Φ** is the **flux linking** with the search coil and
- **i** is the **instantaneous current** flowing in the circuit.
- **EMF induced in Search coil** due to change in linking flux at any instant, **e** is

$$e = N \frac{d\Phi}{dt}$$

$\frac{d\Phi}{dt}$ is the **rate of change of flux linking** with the search coil

THEORY OF FLUX METER

- The EMF induced in the **flux meter coil** due to the movement of coil in the permanent magnet field e_b ,

$$e_b = K \frac{d\theta}{dt}$$

- **K** - constant of flux meter, depends upon the dimensions of instrument
- The **EMF** due to inductance in the circuit = $L \frac{di}{dt}$
- **Voltage drop** in the resistance of the circuit = Ri
- Equation connecting all is

$$e = e_b + L \frac{di}{dt} + Ri$$

- Since **resistance is small** Ri can be neglected.

THEORY OF FLUX METER

$$e = e_b + L \frac{di}{dt} + R i$$

$$N \frac{d\Phi}{dt} = K \frac{d\theta}{dt} + L \frac{di}{dt}$$

Integrating the above eqn over a time **0 to T**, where T is the time taken for the **flux change**

$$\int_0^T N \frac{d\Phi}{dt} dt = \int_0^T K \frac{d\theta}{dt} dt + \int_0^T L \frac{di}{dt} dt$$

Assuming Φ_1 and Φ_2 be the **fluxes linking** with the search coil, θ_{i_0} and θ_2 the **deflections** of the fluxmeter and i_1 and i_2 the **currents flowing** through the search coil circuit at the beginning and at the end of the flux change respectively then the expression may be **re-written** with proper limits as

THEORY OF FLUX METER

- Since current i_1 and i_2 therefore $\int_{i_1}^{i_2} L di = 0$

$$N(\Phi_2 - \Phi_1) = K(\theta_2 - \theta_1)$$

OR

$$(\theta_2 - \theta_1) = \frac{N}{K}(\Phi_2 - \Phi_1)$$

- i.e. deflection $(\theta_2 - \theta_1)$ is proportional to the change in flux-turns.
- The flux meter will therefore have a uniform scale.

ADVANTAGES & DISADVANTAGES OF FLUX METER

- Advantages of Fluxmeter
 - The fluxmeter is portable.
 - The scale of the fluxmeter is calibrated in Weber meters.
 - The deflection of the coil is free from the time taken by the flux to change.
- Disadvantages
 - The only disadvantage of the fluxmeter is that it is less sensitive and accurate

MEASUREMENT OF FLUX DENSITY

- The measurement of flux density inside a specimen can be done by winding a search coil over the specimen. This search coil is known as a "*B coil*".
- This search coil is then connected to a ballistic galvanometer or a flux meter.
- Let us consider that we have to measure the flux density in a ring specimen shown in the Figure.
- The ring specimen is wound with a magnetizing winding which carries a current I .
- A search coil of convenient number of turns is wound on the specimen and connected through a resistance and calibrating coil, to a ballistic galvanometer as shown.

MEASUREMENT OF FLUX DENSITY

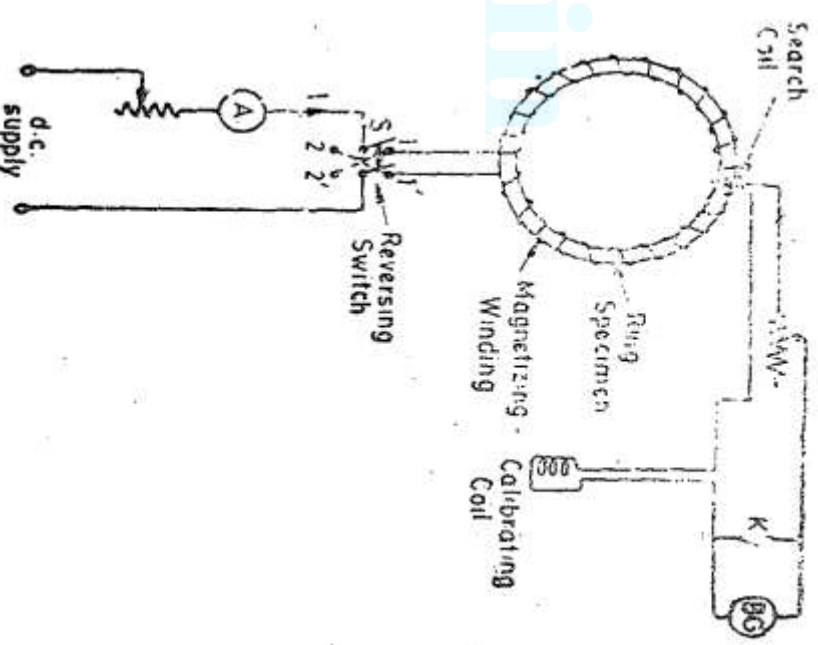
- The current through the magnetizing coil is reversed and therefore the flux linkages of the search coil changes inducing an emf in it.
- This emf sends a current through the ballistic galvanometer causing it to deflect.

- Let Φ = flux linking the search coil,

R = resistance of the ballistic galvanometer circuit

N = number of turns in the search coil,

and t = time taken to reverse the flux.



MEASUREMENT OF FLUX DENSITY

∴ Average emf induced in the search coil

$$e = N \frac{d\Phi}{dt} = \frac{2N\Phi}{t}$$

Average current through the ballistic galvanometer is $i = \frac{2N\Phi}{Rt}$.

Charge passing $Q = it = \frac{2N\Phi}{R}$.

- Let θ_1 be the throw of the galvanometer and Kq be the constant of the galvanometer expressed in coulomb per unit deflection.

MEASUREMENT OF FLUX DENSITY

Charge indicated by ballistic galvanometer

$$= K_q \theta_1.$$

$$\therefore \frac{2N\Phi}{R} = K_q \theta_1.$$

$$\text{or flux } \Phi = \frac{RK_q \theta_1}{2N}$$

- In a uniform field and with search coil turns at right angles to the flux density vector, we have, flux density,

$$B = \frac{\text{flux}}{\text{area}} = \frac{\Phi}{A_s} = \frac{RK_q \theta_1}{2NA_s}$$

where A_s = cross-sectional area of specimen.

- Thus the flux density may be measured by a ballistic galvanometer by observing its throw .

MEASUREMENT OF FLUX DENSITY

- Correction for air flux
 - Flux linking with the search coil is the sum of the flux existing in the specimen and the flux which is present in the air space between the specimen and the search coil.
 - Observed value of flux = true value of flux in the specimen + flux in the air space between specimen and search coil.
- $$B'As = BAs + \mu_o H (A_c - A_s)$$
- Hence true value of flux density

$$B = B' - \mu_o H \left(\frac{A_c}{A_s} - 1 \right)$$

B' = observed value of flux density Wb/sq m

B = true value of flux density in specimen Wb/sq m

A_s = area of cross section of specimen, sq m

A_c = area of cross section of coil, sq m

DETERMINATION OF B-H CURVE

- 1) Method of reversal
 - A ring shaped specimen whose dimensions are known is used for the purpose.
 - The test circuit is same used for flux density measurement.
 - Ring specimen is prepared by placing search coil and magnetizing coil with proper insulation.
 - After demagnetizing, the test is started by setting the magnetizing current to its lowest value.
 - With galvanometer key K closed, the iron specimen is then brought into a 'reproducible cyclic magnetic state' by throwing the reversing switch S backward and forward about twenty times.

DETERMINATION OF B-H CURVE

- Key K is now opened and the value of flux corresponding to this value of H is measured by reversing the switch S and noting the throw of galvanometer.
- The value of flux density corresponding to this H can be calculated by dividing the flux by the area of the specimen.
- The above procedure is repeated for various values of H upto the maximum testing point.
- The B-H curve may be plotted from the measured values of B corresponding to the various values of H

DETERMINATION OF B-H CURVE

2) Step by step method.

- The circuit for this test is shown in figure below
- The magnetizing winding is supplied through a potential divider having a large number of tapings .
- The tapplings are arranged so that the magnetizing force H may be increased, in a number of suitable steps, upto the desired maximum value.
- The specimen before being tested is demagnetized .

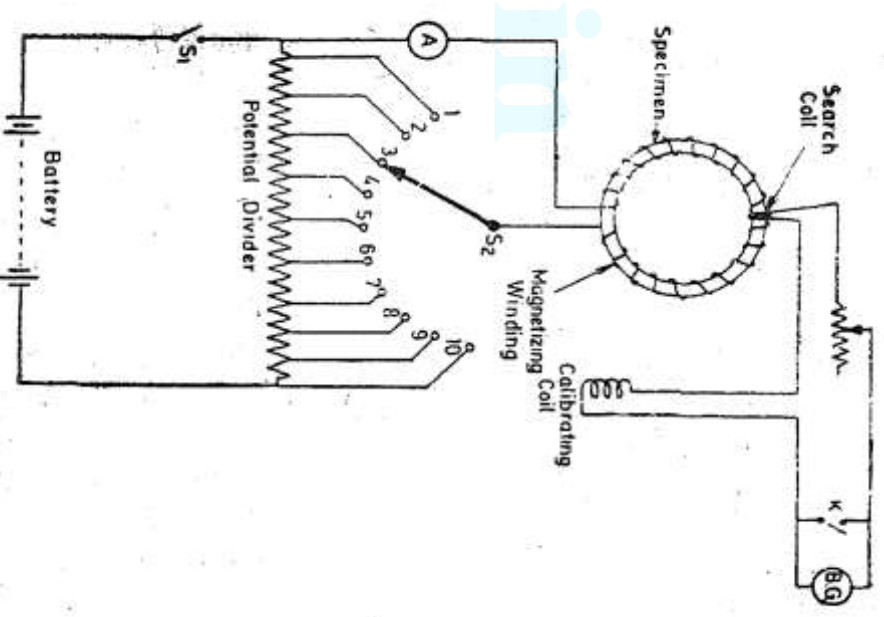


Fig. 18'4. Determination of B - H curve by step by step method.

DETERMINATION OF B-H CURVE

- The tapping switch S_2 is set on tapping 1 and the switch S_1 is closed.
- The throw of the galvanometer corresponding to this increase in flux density in the specimen, from zero to some value B_1 .
- The value of B_1 can be calculated from the throw of the galvanometer.
- The value of corresponding magnetising force H_1 may be calculated from the value of current flowing in the magnetising winding at tapping 1.

DETERMINATION OF B-H CURVE

- Magnetising force is then increased to H_2 by switching S_2 suddenly to tapping 2, and the corresponding increase in flux density ΔB is determined from the throw of the galvanometer.
- Then flux density B_2 corresponding to magnetising force H_2 is given by $B_1 + \Delta B$
- This process is repeated for other values of H upto the maximum point, and the complete B-H curve is obtained.

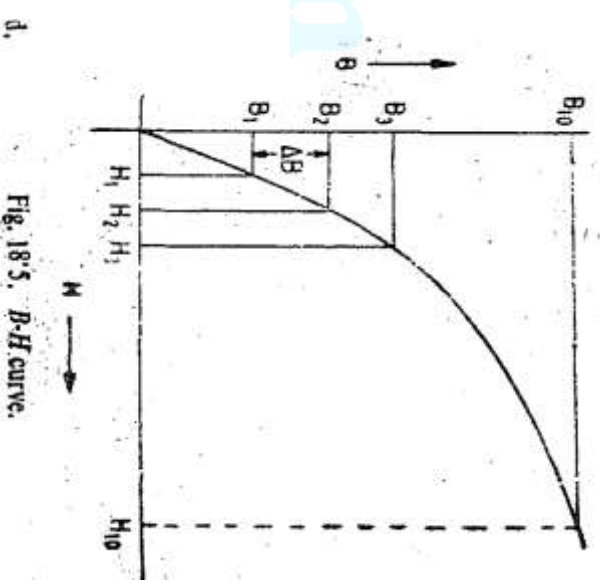


Fig. 18'5. *B-H* curve.

PROBLEM

An iron ring of 350 mm^2 cross sectional area and length 1 m is wound with magnetizing winding of 100 turns. A search coil of 200 turns is connected to a BG having a constant of 1 mC per scale division. The total resistance of search coil is 2000Ω . On reversing the current of 10 A in the magnetizing winding, the galvanometer shows a deflection of 100 scale division. Calculate, the flux density and permeability.

$$Q = e f = \frac{2 N \phi}{R} \quad \begin{array}{l} l = 1 \text{ m} \\ K_g = 1 \times 10^{-6} \end{array}$$

$$\Theta = 100$$

$$Q = K_g \Theta \quad N_s = 200$$

$$Q = \frac{2 N \phi}{R} = K_g \Theta \quad \begin{array}{l} R = 2000 \Omega \\ A = 350 \text{ mm}^2 \\ N_{\text{specimen}} = 100 \end{array}$$

$$\phi = \frac{K_g \Theta R}{2 N} = \frac{1 \times 10^{-6} \times 100 \times 2000}{2 \times 200}$$

$$= 500 \text{ mWb}$$

PROBLEM

$$B = \phi / A = \frac{500 \times 10^{-6}}{350 \times 10^{-6}} = 1.429 \text{ Wb/m}^2$$

$$\mu = \frac{B}{H} = \frac{B}{\frac{NI}{l}} = \frac{1.429 \times 1}{100 \times 10} \\ = \underline{\underline{1.429 \times 10^{-3}}}$$

HYSTERESIS LOOP

- Hysteresis loop is a four quadrant B-H graph from where the hysteresis loss, coercive force and retentively of magnetic material are obtained.
- To understand hysteresis loop, we suppose to take a magnetic material to use as a core around which insulated wire is wound. The coils is connected to the supply (DC) through variable resistor to vary the current I . We know that current I is directly proportional to the value of magnetizing force (H).

$$H = NI/l$$

Where, N = no. of turn of coil and l is the effective length of the coil. The magnetic flux density of this core is B which is directly proportional to magnetizing force H

HYSTERESIS LOOP

- Step 1:
When supply current $I = 0$, so no existence of flux density (B) and magnetizing force (H). The corresponding point is 'O' in the graph above.
- Step 2:
When current is increased from zero value to a certain value, magnetizing force (H) and flux density (B) both are set up and increased following the path $0 - a$.
- Step 3:
For a certain value of current, flux density (B) becomes maximum ($B_{\max}$). The point indicates the magnetic saturation or maximum flux density of this core material. All element of core material get aligned perfectly. Hence $H_{\max}$ is marked on H axis. So no change of value of B with further increment of H occurs beyond point 'a'.

HYSTERESIS LOOP

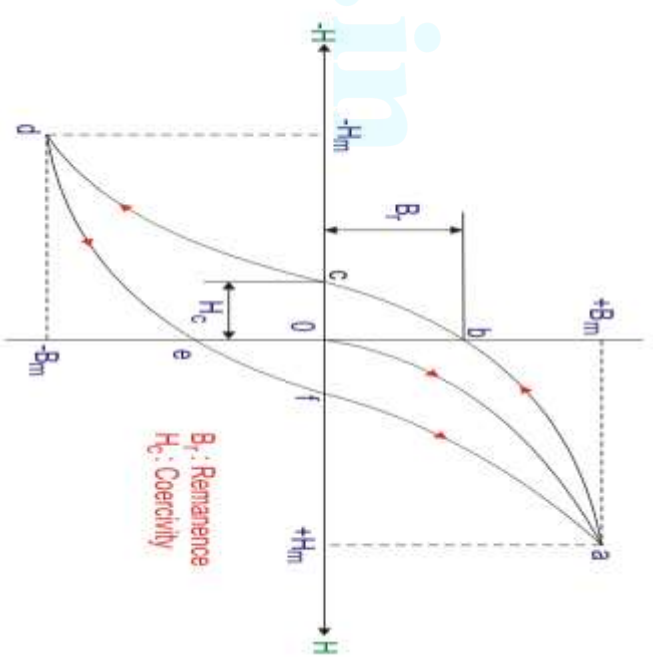
- Step4:
When the value of current is decreased from its value of magnetic flux saturation, H is decreased along with decrement of B not following the previous path rather following the curve $a - b$.
- Step5:
The point 'b' indicates $H = 0$ for $I = 0$ with a certain value of B . This lagging of B behind H is called hysteresis. The point 'b' explains that after removing of magnetizing force (H), magnetism property with little value remains in this magnetic material and it is known as residual magnetism (B_r). Here $o - b$ is the value of residual flux density due to retentivity of the material.

HYSTERESIS LOOP

- Step 6:
If the direction of the current I is reversed, the direction of H also gets reversed. The increment of H in reverse direction following path $b - c$ decreases the value of residual magnetism (B_r) that gets zero at point 'c' with certain negative value of H . This negative value of H is called coercive force (H_c)
- Step 7:
 H is increased more in negative direction further; B gets reverses following path $c - d$. At point 'd', again magnetic saturation takes place but in opposite direction with respect to previous case. At point 'd', B and H get maximum values in reverse direction, i.e. ($-B_m$ and $-H_m$).
- Step 8:
If we decrease the value of H in this direction, again B decreases following the path de . At point 'e', H gets zero valued but B is with finite value. The point 'e' stands for residual magnetism ($-B_r$) of the magnetic core material in opposite direction with respect to previous case.

HYSTERESIS LOOP

- Step 9:
If the direction of H again reversed by reversing the current I , then residual magnetism or residual flux density ($-B_r$) again decreases and gets zero at point 'f' following the path $e - f$. Again further increment of H , the value of B increases from zero to its maximum value or saturation level at point a following path $f - a$.
- The path $a - b - c - d - e - f - a$ forms hysteresis loop.
- The shape and the size of the hysteresis loop depend on the nature of the material chosen



HYSTERESIS LOOP

- **Residual flux density** is the certain value of magnetic flux per unit area that remains in the magnetic material without presence of magnetizing force (i.e. $H = 0$).
- **Coercive force** is defined as the negative value of magnetizing force ($-H$) that reduces residual flux density of a material to zero.
- Retentivity is defined as the degree to which a magnetic material gains its magnetism after magnetizing force (H) is reduced to zero.

DETERMINATION OF HYSTERESIS LOOP

➤ Step by step method.

- The determination of hysteresis loop by this method is done by simply continuing the procedure just described for the determination of B-H curve using step by step method.
- After reaching the point of maximum H i.e., when switch S₂ is at tapping 10, the magnetising current is next reduced, in steps to zero by moving switch S₂ down through the tapping points 9, 8, 7 3, 2, 1

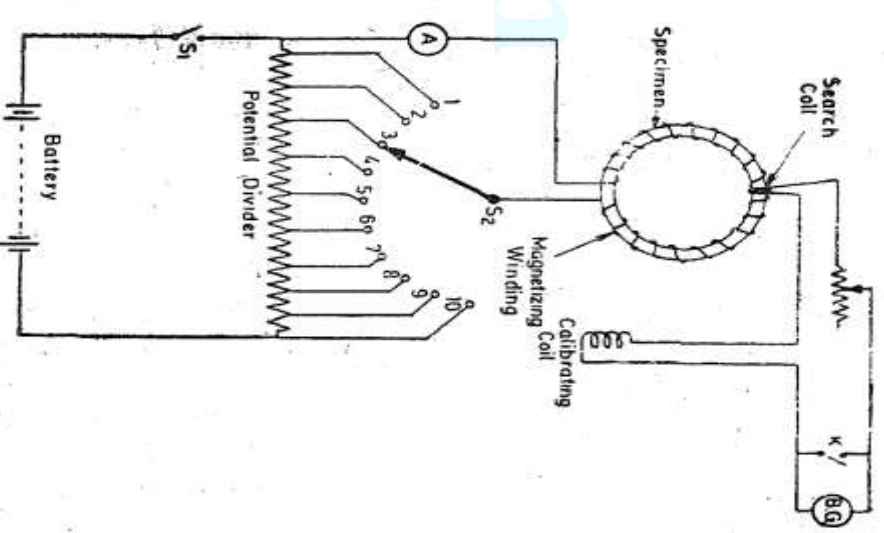


Fig. 18'4. Determination of B-H curve by step by step method.

DETERMINATION OF HYSTERESIS LOOP

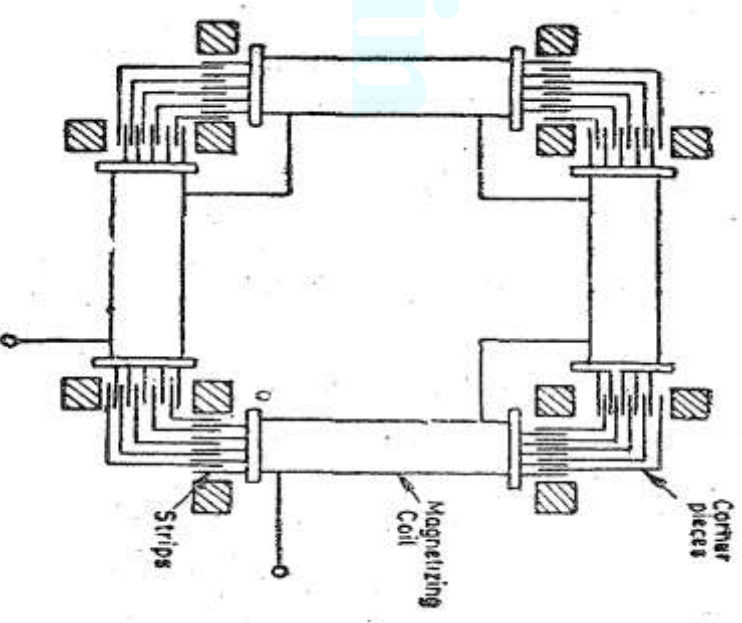
- After reduction of magnetising force to zero, negative values of H are obtained by reversing the supply to potential divider and then moving the switch S2 up again in order 1, 2, 3, 7, 8, 9, 10.

LLOYD-FISHER SQUARE FOR IRON LOSS MEASUREMENT

- This is the most commonly used magnetic square.
- The strips used are usually 0.25 m long and 50 to 60 mm wide.
- These strips are built up into four stacks.
- The stacks or strips are placed inside four similar magnetising coils of large cross-sectional area.
- These four coils are connected in series to form the primary winding..
- Each magnetising coil has two similar single layer coils underneath it. They are called secondary coils.

LLOYD-FISHER SQUARE FOR IRON LOSS MEASUREMENT

- Thus in a magnetic square there are eight secondary coils. These secondary coils are connected in series in groups of four, one from each core, to form two separate secondary windings.
- The ends of the strips project beyond the magnetising coils.
- The magnetic circuit is completed by bringing the four stacks together in the form of a square and joining them at the corners.



LLOYD-FISHER SQUARE FOR IRON LOSS MEASUREMENT

- The corner joints are made by a set of standard right angled corner pieces. The corner pieces are of the same material as strips or at least a material having the same magnetic properties.

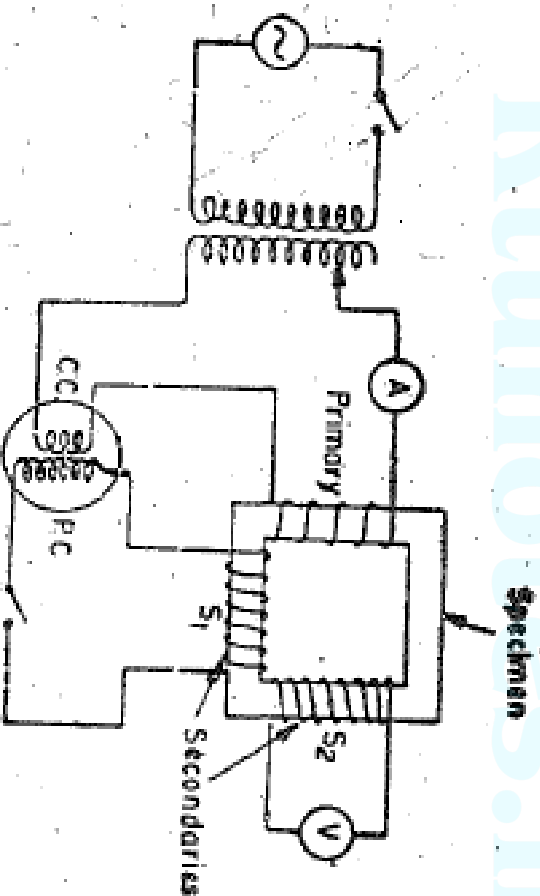


Fig. 18'21. Wattmeter method for determining iron loss.

LLOYD-FISHER SQUARE FOR IRON LOSS MEASUREMENT

Theory. The electrostatic voltmeter connected across secondary winding S_2 measures the rms value of induced voltage. This value of induced voltage is

$$E = 4 K_f \Phi_m f N_2 = 4 K_f B_m A_s f N_2$$

where K_f = form factor (1.11 for sinusoidal waves),

Φ_m = maximum value of flux ; Wb,

B_m = maximum value of flux density ; Wb/m²,

A_s = cross-sectional area of specimen ; m²,

f = frequency ; Hz,

N_2 = number of turns in the secondary winding S_2 .

and

$$\therefore \text{Maximum value of flux density } B_m = \frac{E}{4 K_f A_s f N_2}$$

LLOYD-FISHER SQUARE FOR IRON LOSS MEASUREMENT

Correction must be applied to this value of flux density as the secondary winding S_2 encloses the flux in the air space between specimen and the coil in addition to the flux in the specimen, as discussed earlier on page 571.

The actual value of flux density in the specimen (Eqn. 18'2) is

$$B_m = B_m' - \mu_0 H_m \left(\frac{A_s}{A_o} - 1 \right) \quad \dots (18'22)$$

where A_o = cross sectional area of coil.
and H_m = magnetising force corresponding to maximum flux density. This may be taken from the B - H curve of the specimen.

The wattmeter reading includes both the iron loss in the specimen and the copper loss in the secondary winding circuit. The copper loss in the secondary winding can be calculated and subtracted from the wattmeter readings.

Let P_t = total iron loss, P = wattmeter reading,

V = voltage applied to wattmeter pressure coil,

E = voltmeter reading = voltage induced in S_2 ,

LLOYD-FISHER SQUARE FOR IRON LOSS MEASUREMENT

r_p = resistance of wattmeter pressure coil, r_e = resistance of coil S_1 ,

and I_p = current in the pressure coil circuit.

Now, voltage induced in S_1 is equal to the voltage induced in S_2 since both of them have equal number of turns and they link with the same flux.

$\therefore$ Voltage induced in $S_1 = E$

The leakage reactances of winding S_1 and that of pressure coil are very small as compared with resistance of the pressure coil circuit. If we neglect the leakage reactances of winding S_1 and pressure coil, we have

$$E = I_p(r_p + r_e).$$

$\therefore$ Total iron loss in specimen + total copper loss in the secondary winding circuit = PE/V

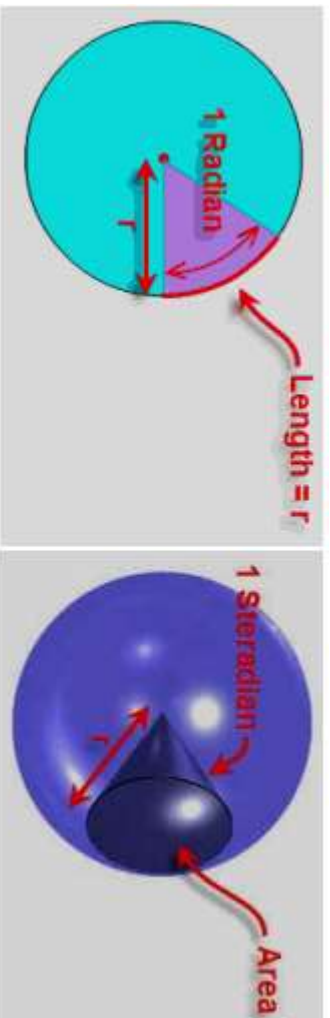
Total copper loss in the secondary winding circuit = $E^2/r_p + r_e$.

$\therefore$ Total iron loss in the specimen $P_i = P \frac{E}{V} - \frac{E^2}{r_p + r_e} = P \left(1 + \frac{r_e}{r_p} \right) - \frac{E^2}{r_p + r_e}$... (18.23)

Specific iron loss i.e., iron loss per kg can be calculated by dividing the total iron loss by the weight of the specimen.

DEFINITIONS

- Luminous flux
 - It is the light energy radiated out per second from the body in the form of luminous light waves. It is thus the rate of energy radiation in the form of light. Its unit is lumen
- Luminous intensity
 - Luminous intensity or Candle-power of a point source in a given direction is the luminous flux (number of lumens) radiated out per unit solid angle.



DEFINITIONS

- Illumination (E) or Illuminance:
 - When the luminous flux falls on a surface, it is said to be illuminated. Illumination is the luminous flux received by a surface per unit area. Its unit is Lux.

MEASUREMENT OF LUMINOUS INTENSITY

➤ Photoelectric device

- A photoelectric device is a piece of apparatus which allows the flow of current only when light falls on the device.
- Also the magnitude of the current varies with the intensity of light and hence these devices can be used for photometric measurements.
- The different photo-electric devices are,
 - (i) Photo emissive cells
 - (ii) Photo conductive cells
 - (iii) Photo voltaic cells

PHOTOCONDUCTIVE CELL

- It is a photo conductive transducer
- A photo conductive transducer is one in which the change in conductivity of a semi-conducting material occurs with change in radiation intensity.
- The change in conductivity appears as change in resistance and therefore these devices are photo resistive cells.
- Thus from the point of view of transduction the resistance changes with light intensity.
- Photo conductive cells is a photo conductive transducer.

PHOTOCONDUCTIVE CELL

- The principle of working is,
 - In a semi-conductor an energy gap exists between conduction electrons and valence electrons.
 - In a semi-conductor photoconductive transducer, a photon is absorbed and thereby excites an electron from valence band to conduction band.
 - As electrons are excited from valence band to conduction band, the resistance decreases, making the resistance an inverse function of radiation intensity.

PHOTOCONDUCTIVE CELL

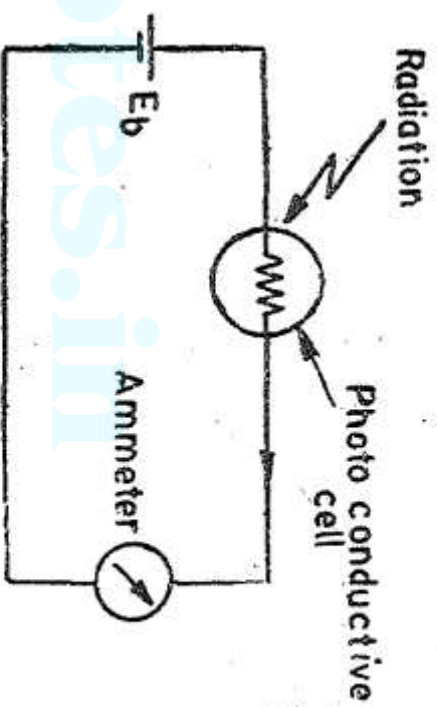
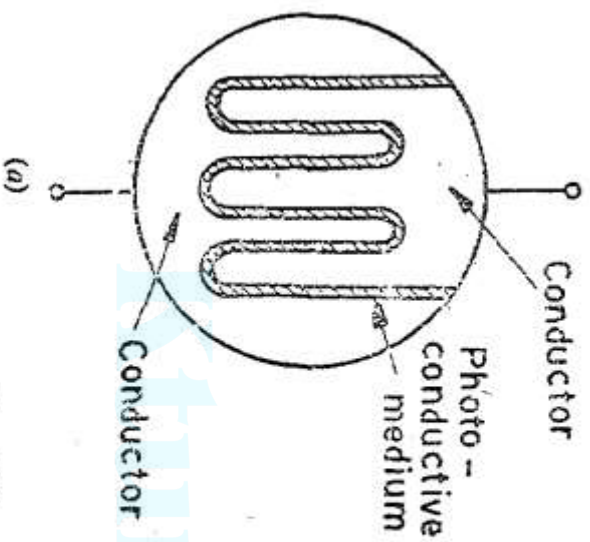


Fig. 19'26. Photoconductive cell and its circuit.



PHOTOCONDUCTIVE CELL

- The most two commonly used photoconductive semiconductor materials are cadmium sulphide (CdS) with a band gap of 2.42 eV and cadmium selenide (CdSe) with a 1.74 eV band gap.
- On account of these large energy band gaps, both the materials have a very high resistivity at ambient temperature.
- So a special kind of construction has to be used, which minimises resistance while providing maximum surface.
- This special type of construction is shown in Fig (a). This construction gives minimum length and maximum area.
- The photoconductive material is deposited on a ceramic substrate in a zig zag manner. The electrodes are of tin or indium

PHOTOCONDUCTIVE CELL

- When the cell is kept in darkness, its resistance is called dark resistance.
- The dark resistance may be very high.
- If the cell is illuminated its resistance decreases.
- The resistance depends on the physical character of the photoconductive layer as well as on the dimensions of the cell and its geometric configuration.
- The current depends upon the d.c. voltage applied.
- When the illumination level changes, the conductivity of photoconductive cell changes and thus the current through the ammeter also changes. The change in current is an indication of luminous intensity
- The photoconductive cell suffers from a major disadvantage that temperature changes cause substantial resistance changes for a particular light intensity.

PHOTOVOLTAIC CELL

- This is an important class of photo detectors.
- They generate a voltage which is proportional to EM radiation intensity.
- They are called photovoltaic cells because of their voltage generating characteristics.
- They in fact convert the EM energy into electrical energy.
- They are active transducers. i e., they do not need an external source to power them.

PHOTOVOLTAIC CELL

- The cell is a giant diode with a p-n junction between appropriately doped semiconductors.
- Photons striking the cell pass through the thin p-doped upper layer and are absorbed by electrons in the lower n layer, causing formation of conduction electrons and holes.
- The depletion zone potential of the p-n junction then separates these conduction holes and electrons causing a difference of potential to develop across the junction .

PHOTOVOLTAIC CELL

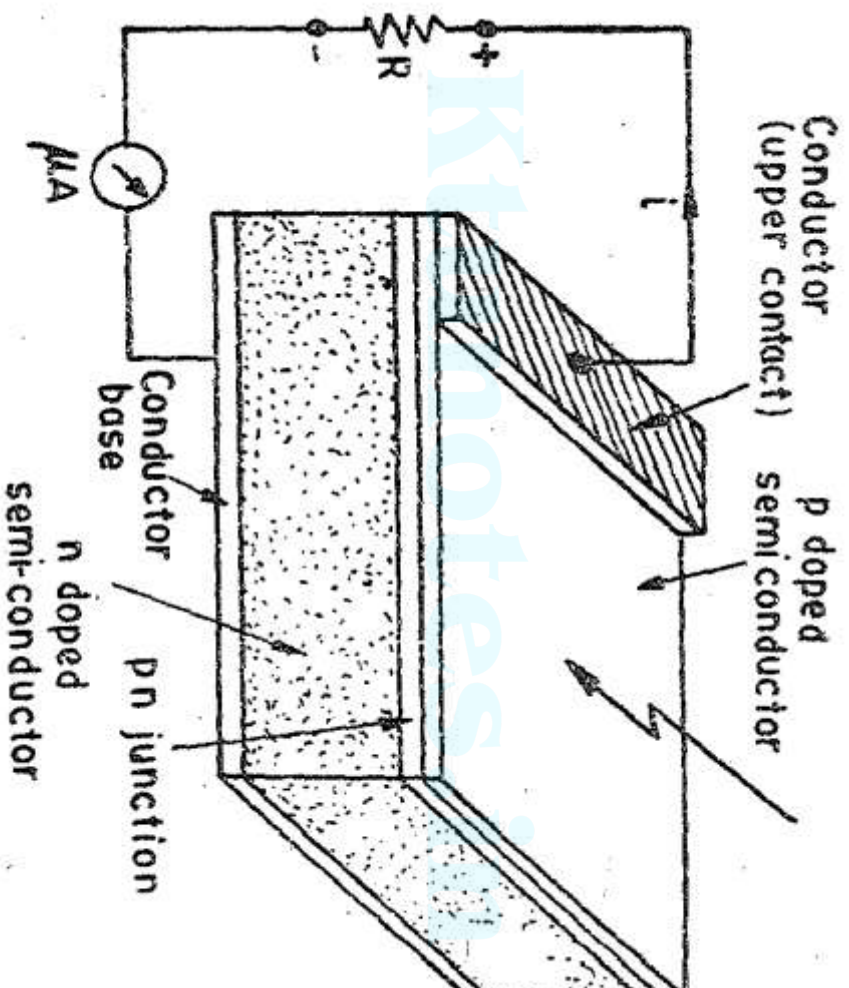


Fig. 10.99 Photovoltaic cell

PHOTOVOLTAIC CELL

- The current verses luminous flux characteristics of a typical photo voltaic cell is shown here.
- It shows that the current is a function of the cell incident luminous flux and hence micro ammeter shown in Fig. may be directly calibrated to read luminous flux or luminous intensity.
- The photovoltaic cells can operate satisfactorily in the temperature range of -100 to 125°C

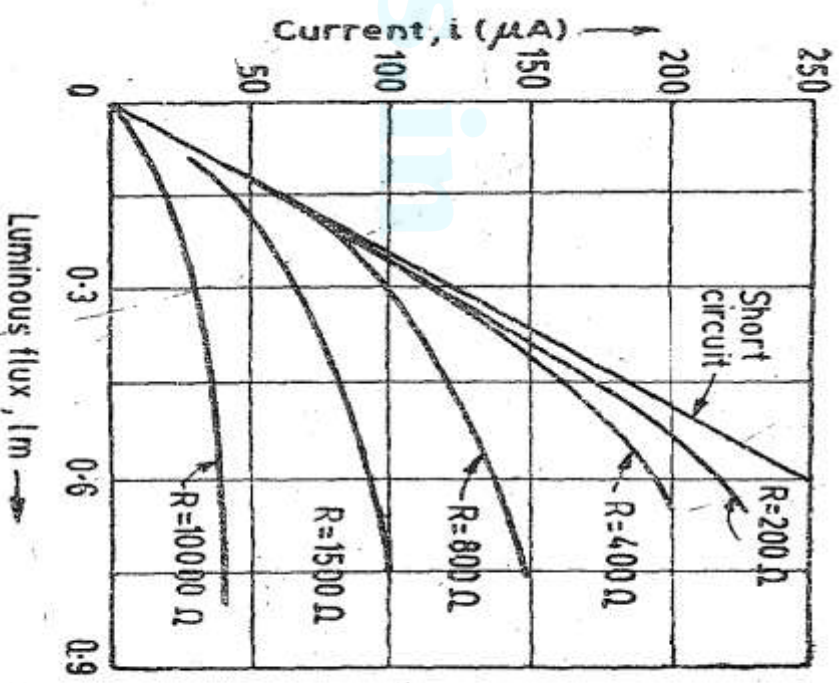


Fig. 19-33. Current versus luminous flux characteristics for a photovoltaic cell.

MEASUREMENT OF TEMPERATURE

- **Temperature Measuring Devices**
 - Resistance temperature detectors/Resistance thermometers
 - Semiconductor thermometers
 - Thermistors
 - Thermocouples
 - Radiation pyrometers

RESISTANCE TEMPERATURE DETECTOR

- A Resistance Temperature Detector (also known as a Resistance Thermometer or RTD) is an electronic device used to determine the temperature by measuring the resistance of an electrical wire.
- This wire is referred to as a temperature sensor.
- If we want to measure temperature with high accuracy, an RTD is the ideal solution.
- The resistance of a conductor changes when its temperature is changed. This property is utilized for measurement of temperature.

RESISTANCE TEMPERATURE DETECTOR

- The variation of resistance R with temperature T can be represented by the following relationship for most of the metals as

$$R = R_0(1 + \alpha_1 T + \alpha_2 T^2 + \dots + \alpha_n T^n + \dots)$$

R_0 = resistance at temperature $T = 0$ and $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ are constants.

- The requirements of a conductor material to be used in these thermometers are,
 - The change in resistance of material per unit change in temperature should be as large as possible
 - The resistance of the materials should have a continuous and stable relationship with temperature.

RESISTANCE TEMPERATURE DETECTOR

- The characteristics of various materials used for resistance thermometers are plotted

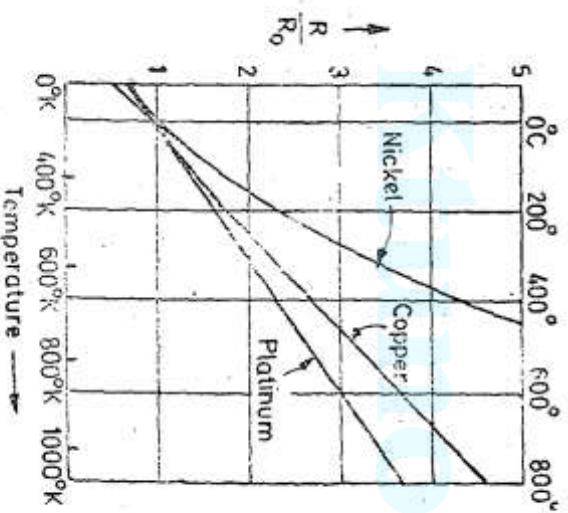
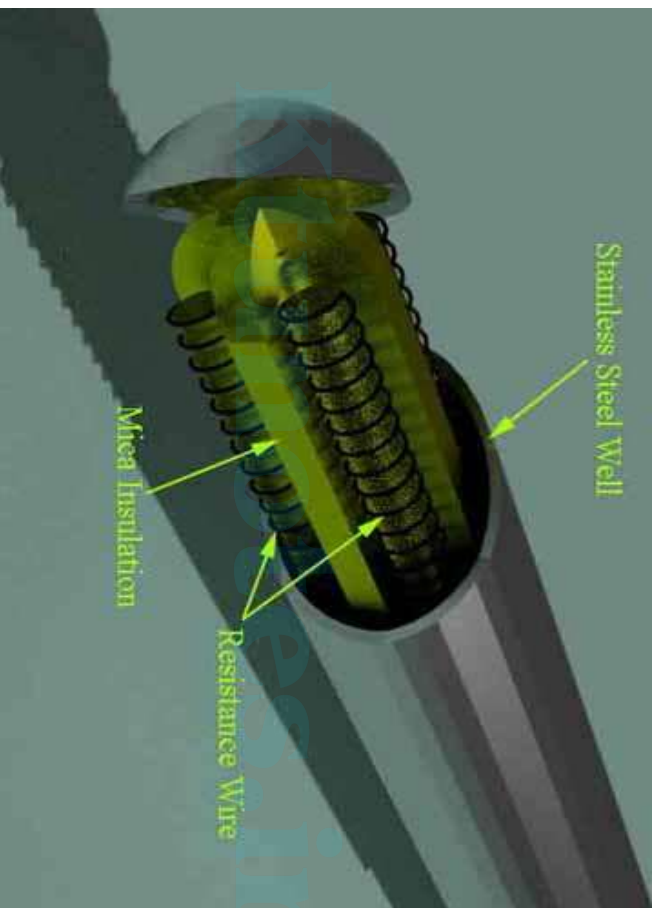


Fig. 25-20. Characteristics of materials used for resistance thermometers.

RESISTANCE TEMPERATURE DETECTOR



- The construction is typically such that the wire is wound on a form (in a coil) on notched mica cross frame to achieve small size, improving the thermal conductivity to decrease the response time and a high rate of heat transfer is obtained.

RESISTANCE TEMPERATURE DETECTOR

- In the industrial RTD's, the coil is protected by a stainless steel sheath or a protective tube so that, the physical strain is negligible .
- If the strain on the wire is increasing, then the tension increases. Due to that, the resistance of the wire will change which is undesirable.
- Mica is placed in between the steel sheath and resistance wire for better electrical insulation.
- RTD value is measured by using a bridge circuit.

RESISTANCE TEMPERATURE DETECTOR

- By supplying the constant electric current to the bridge circuit and measuring the resulting voltage drop across the resistor, the RTD resistance can be calculated. Thereby, the temperature can be also determined.
- This temperature is determined by converting the RTD resistance value using a calibration expression.

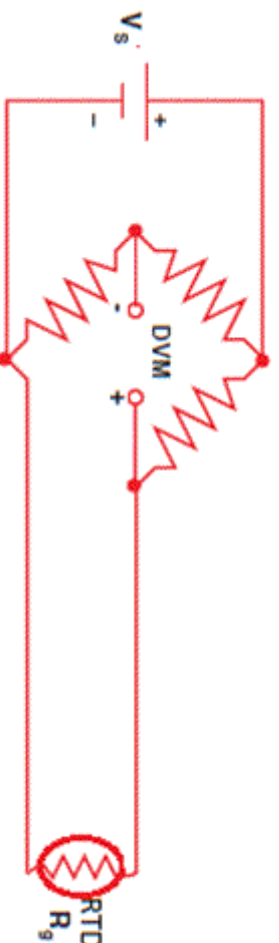


Fig.3. Two wires RTD Bridge

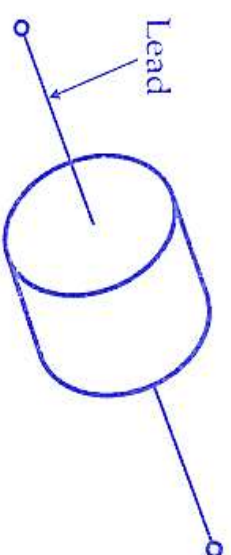
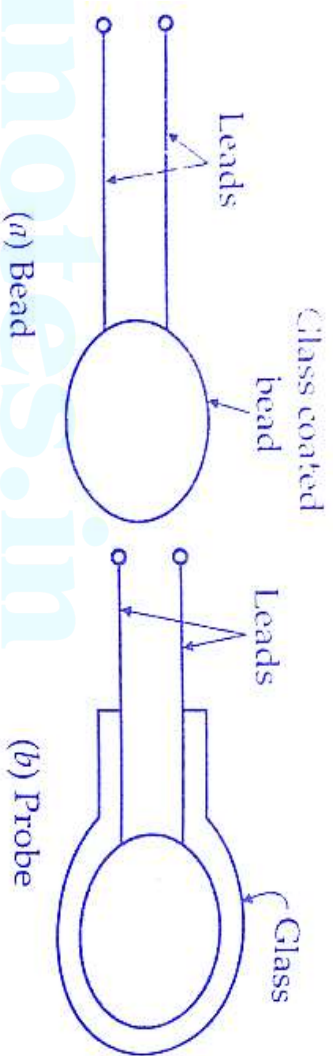
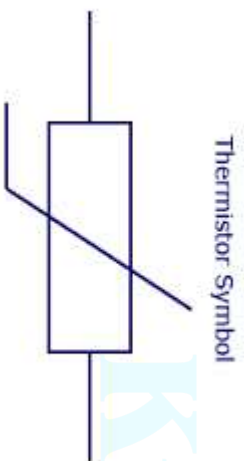
THERMISTORS

- Thermistors is the contraction of term “Thermal Resistors”.
- They are essentially semi-conductors which behave as resistors with a high negative temperature co-efficient of resistance.
- In some cases the resistance of a thermistor at room temperature may decrease as much as *5 per cent for each 1 °C* rise in temperature. This high sensitivity to temperature changes make the thermistors extremely useful for precision temperature measurements, control and compensation.
- Thermistors are widely used in such applications especially in the temperature range of -60°C to $+15^{\circ}\text{C}$. The resistance of thermistors ranges from 0.5Ω to $0.75\text{ M}\Omega$.
- Thermistors are composed of sintered mixture of metallic oxides such as manganese, nickel, cobalt, copper, iron and uranium.

THERMISTORS

- They are available in variety of sizes and shapes .
- The thermistors may be in the form of beads, rods or discs.
- A thermistor in the form of a bead is smallest in size and the bead may have a diameter of 0.015 mm to 1.25 mm.
- Beads may be sealed in the tips of solid glass rods to form probes which may be easier to mount than the beads. Glass probes have a diameter of about *2.5 mm and a length which* varies from 6 mm to 50mm.
- Discs are made by pressing material under high pressure into cylindrical flat shapes with diameters ranging from 2.5 mm to 25 mm.

THERMISTORS



(c) Disc



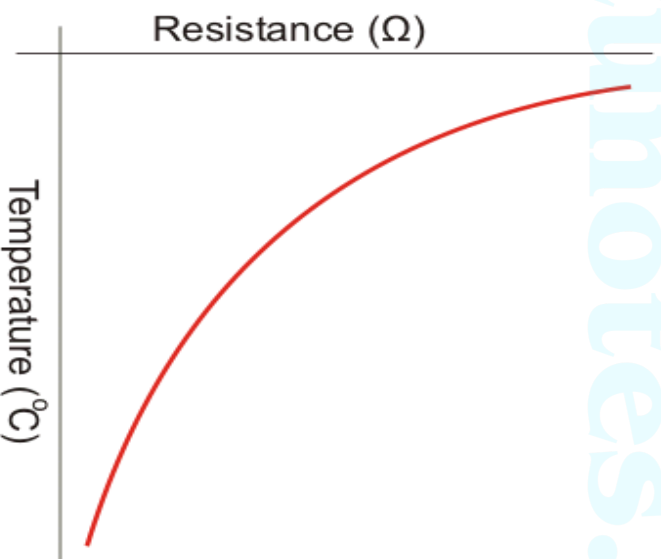
(d) Rod

THERMISTORS

- The working principle of a thermistor is that its resistance is dependent on its temperature. We can measure the resistance of a thermistor using an ohmmeter.
- If we know the exact relationship between how changes in the temperature will affect the resistance of the thermistor – then by measuring the thermistor's resistance we can derive its temperature.
- How much the resistance changes depends on the type of material used in the thermistor.
- The relationship between a thermistor's temperature and resistance is non-linear.

THERMISTORS

- A typical thermistor graph is shown below:
- If we had a thermistor with the above temperature graph, we could simply line up the resistance measured by the ohmmeter with the temperature indicated on the graph.

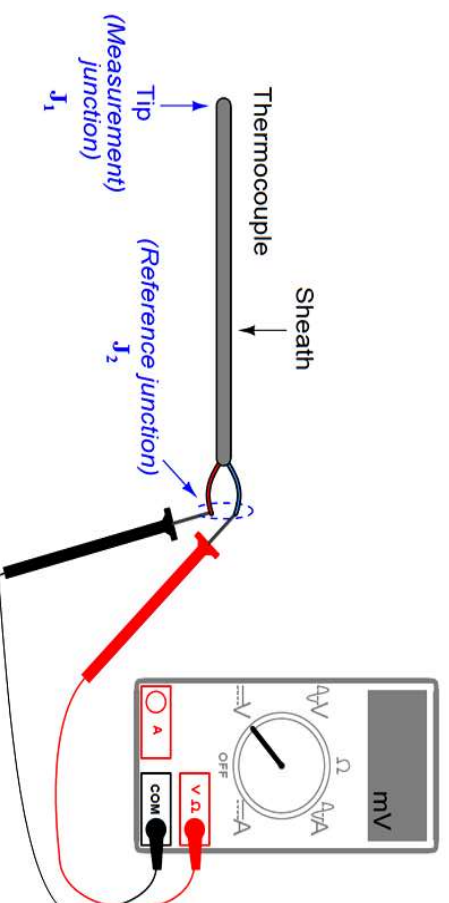


THERMISTORS

- Applications
 - Digital thermometers (thermostats)
 - Automotive applications (to measure oil and coolant temperatures in cars & trucks)
 - Household appliances (like microwaves, fridges, and ovens)
 - Circuit protection
 - Rechargeable batteries (ensure the correct battery temperature is maintained)
 - To measure the thermal conductivity of electrical materials
 - Temperature compensation

THERMOCOUPLE

- Thermocouple, also called thermal junction, thermoelectric thermometer, or thermel.
- It is a temperature-measuring device consisting of two wires of different metals joined at each end.
- One junction is placed where the temperature is to be measured, and the other is kept at a constant lower temperature.



THERMOCOUPLE

- A measuring instrument is connected in the circuit. The temperature difference causes the development of an electromotive force (known as the Seebeck effect) that is approximately proportional to the difference between the temperatures of the two junctions.
- Temperature can be read from standard tables, or the measuring instrument can be calibrated to read temperature directly.
- Any two different metals or metal alloys exhibiting the thermoelectric effect, are used as thermocouples e.g., antimony and bismuth, copper and iron, or copper and constantan (a copper-nickel alloy).

THERMOCOUPLE

- Usually platinum, either with rhodium or a platinum-rhodium alloy, is used in high-temperature thermocouples.
- The most common type is nickel-aluminum and nickel-chromium wires because of its wide temperature range .